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Journal Title: Archives internationales de pharmacodynamie et de the?rapie

Volume: 327 **Issue:** 3

Month/Year: May 1994**Pages:** 266-78

Article Author: Chen T, Kato H, Ban H, Nakata N, Liu XH, Itoyama Y, Kogure K

Article Title: Protective effects of a novel calcium antagonist with platelet-activating factor-antagonistic action

Imprint:

ILL Number: 54423813



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Protective Effects of a Novel Calcium Antagonist with Platelet-Activating Factor-Antagonistic Action, F-0401, against Ischemic Brain Damage

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Abstract—The protective effects of a novel dihydropyridine calcium antagonist with platelet-activating factor-antagonistic action, F-0401, on ischemic brain damage were investigated using experimental ischemia models in rats and gerbils.

F-0401 (1 and 10 mg/kg, i.p.) prevented increases in water content, determined by the wet-dry method, in ischemic areas 24 hr after 1 hr of middle cerebral artery occlusion in the rat.

Pretreatment with F-0401 (1 and 10 mg/kg, i.p.) prevented extravasation of Evans blue dye in the brain following 2 hr of bilateral carotid artery occlusion and 2 hr of reperfusion in the rat.

Pretreatment with F-0401 (1 and 10 mg/kg, i.p.) protected against neuronal damage to hippocampal CA1 pyramidal cells following 3 and 5 min of forebrain ischemia in the gerbil.

Immunostaining against microtubule-associated protein-2 also demonstrated preservation of CA1 neurons in F-0401-treated animals.

Thus, this study shows that F-0401 prevents the occurrence of brain edema, disruption of blood-brain barrier and neuronal damage caused by cerebral ischemia. The results demonstrate that F-0401 may be a powerful candidate as a therapeutic agent in the treatment of acute stroke in man.

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Introduction

Stroke is one of the common and major causes of disability and intellectual impairment in the elderly. The mechanisms of ischemic brain damage, however, have not been fully elucidated. One of the serious complications of stroke is the formation of brain edema, an increase in water content and tissue volume. Ischemic brain edema has been classified into two types, cytotoxic edema and vasogenic edema, the latter accompanying the disruption of the blood-brain barrier (BBB) (Klatzo, 1967). Another intriguing phenomenon is the delayed neuronal death following transient cerebral ischemia (Kirino, 1982). Pyramidal neurons in the CA1 subfield of the hippocampus, which are most vulnerable to ischemia, are destroyed 3–4 days after transient cerebral ischemia. Although the mechanism of the delayed neuronal death remains elusive, this model has often been used to examine the protective effects of pharmacological interventions.

Strategies to protect the brain from ischemia can be multiple. One of them is to increase the cerebral blood flow and to prevent the disruption of ionic homeostasis, especially the intracellular calcium overload which is believed to lead to cell death (Siesjö and Bengtsson, 1989). Calcium antagonists have been used for this purpose (Abe *et al.*, 1988; Alps *et al.*, 1988; Hara *et al.*, 1990, 1991; Rosenbaum *et al.*, 1991; Sauter and Rudin, 1990; TRUST Study Group, 1990). On the other hand, it has been reported that platelet-activating factor (PAF) is a potent mediator of brain edema that accompanies brain injury including ischemia, and that PAF antagonists ameliorate brain damage resulting from ischemia (Spinnewyn *et al.*, 1987; Lindsberg *et al.*, 1991; Bielenberg *et al.*, 1992). Thus, calcium and PAF antagonists are candidates as therapeutic agents to be used for the treatment of acute stroke in man.

In the present study, we examined the effects of F-0401, a novel dihydropyridine calcium antagonist with PAF-antagonistic action, on postischemic brain edema, disruption of BBB and delayed neuronal death in rats and gerbils. We assumed that it could be more promising to interrupt different pathogenic processes at the same time using a single therapeutic agent. Part of this study has been published in abstract form (Chen *et al.*, 1993).

Methods

Experiment 1

Male Wistar rats (Nippon Clea, Tokyo, Japan), 10–12 weeks old and weighing 240–280 g, were used. They were fasted overnight before surgery and otherwise allowed free access to food and water. The animals were

anesthetized with 2 % halothane in a mixture of 30 % oxygen and 70 % nitrous oxide. The origin of the right middle cerebral artery was occluded with an artificial embolus (a silicone rubber cylinder attached to a nylon surgical thread) introduced via the internal carotid artery, as previously described (Koizumi *et al.*, 1986; Nagasawa and Kogure, 1989). After surgery, anesthesia was discontinued and neurological deficits, such as left hemiparesis with upper limb dominance, were confirmed. Reperfusion of the brain was achieved by pulling the embolus out of the internal carotid artery. Immediately after reperfusion, F-0401 (Fujirebio Inc., Tokyo, Japan), at doses of 1 and 10 mg/kg ($n = 8$ and 13, respectively), or vehicle (3 % DMSO and 2 % HCO-60/ethanol in saline; $n = 13$) was injected i.p. The body temperature of the animals was maintained at close to 37° C during surgery and ischemia using a heating mat and a lamp. Sham-operated animals ($n = 8$) were also included. After 24 hr, the animals were decapitated under deep ether anesthesia and the brains were rapidly removed. Tissue samples of the cortices, supplied by the middle and anterior cerebral arteries (MCA and ACA cortex, respectively) and the caudate-putamen, were dissected out from both hemispheres in a humidified glove box (Fig. 1). The brain tissues were dried at 105° C for 24 hr and the water content was determined by the wet-dry method. Values were expressed as means $\pm$ S.D. Statistical significance was analyzed using the Kruskal-Wallis nonparametric analysis of variance followed by the Williams-Wilcoxon multiple rank sum test.

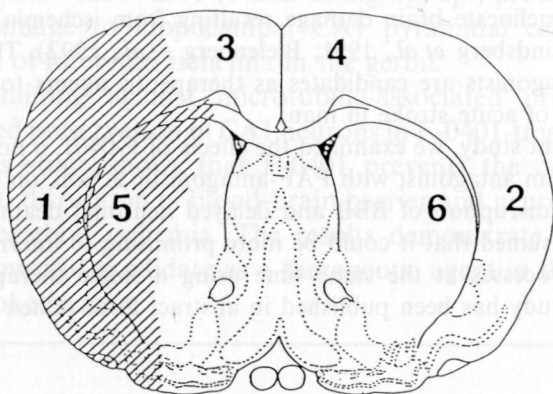


FIG. 1

Anatomic regions of a coronal rat brain section at the level of the caudate-putamen.

The shaded area shows the extent of ischemia (cortex supplied by the right middle cerebral artery and medial segment of the caudate-putamen).

The numbers indicate the regions of tissue sampling for water content measurement: 1 and 2: cortex supplied by the middle cerebral artery; 3 and 4: cortex supplied by the anterior cerebral artery; 5 and 6: caudate-putamen.

Experiment 2

Male Wistar rats (Nippon Clea), 10–11 weeks old and weighing 230–260 g, were used. The animals were anesthetized by inhalation of the same O_2/N_2O /halothane mixture. Bilaterally, the common carotid arteries were occluded for 2 hr, followed by 2 hr of reperfusion. The durations of the ischemia and reperfusion time to produce frequent extravasation of Evans blue dye into the brain were decided in preliminary studies. F-0401 (1 and 10 mg/kg; both $n=6$) or vehicle ($n=8$) was injected i.p. 15 min before occlusion and 5 min before reperfusion. Immediately after reperfusion, Evans blue dye (2 % in saline, 4 ml/kg) was injected i.v. through the jugular vein. After 2 hr, the animals were anesthetized with pentobarbital (50 mg/kg, i.p.) and the brains were perfused with saline through the left cardiac ventricle until colorless perfusion fluid was obtained from the jugular veins, and then the animals were decapitated. The forebrain was cut coronally into three small pieces. A photo was taken from each piece and the degree of extravasation of the dye was semiquantitatively graded as 0 (none), 1 (mild), 2 (moderate) and 3 (severe). The same statistical method as in Experiment 1 was used.

Experiment 3

Male Mongolian gerbils (*Meriones Unguiculatus*, Seiwa Experimental Animals, Fukuoka, Japan), aged 12 weeks and weighing 65–75 g, were used. Anesthesia was induced by the same inhalation as in Experiments 1 and 2. F-0401 (1 and 10 mg/kg) was injected i.p. ($n=10$ and $n=8$, respectively) 15 min before the induction of ischemia. Bilaterally, the common carotid arteries were occluded with aneurysm clips for 3 or 5 min, followed by 7 days of survival. This procedure selectively destroys pyramidal neurons in the hippocampal CA1 subfield (Kirino, 1982; Araki *et al.*, 1989; Kato *et al.*, 1991). Body and temporalis muscle temperatures were maintained at 36.5 to 37° C during ischemia. The body temperature was also monitored at 30 min, 1 hr and 2 hr after ischemia. A similar postischemic hyperthermia was observed in each ischemic group. After 7 days of reperfusion, the brains were perfusion-fixed with 10 % formalin in 0.1 M phosphate buffer following a heparinized saline flush, and then removed. Postfixation of the brains was done in the same fixative until they were embedded in paraffin. Paraffin sections were cut at a thickness of 5 μ m and then stained with cresyl violet. The neuronal density (/mm) of the hippocampal CA1 pyramidal cells was counted under microscope. Statistical analysis was performed in the same way as described above.

The paraffin sections were also immunostained against microtubule-associated protein-2 (MAP2). A monoclonal anti-MAP2 antibody (Amersham, Arlington, IL, U.S.A.) and a Vectastain elite ABC kit (Vector

Laboratories, Burlingame, CA, U.S.A.) were used. The sections were first incubated with 10 % normal serum for 30 min, then with anti-MAP2 antibody (1:1000) overnight at 4° C, followed by incubation with biotinylated secondary antibody (1:200) for 1 hr and with the avidin-biotin-peroxidase complex for 30 min. The color reaction was obtained by incubation with 0.05 % 3,3'-diaminobenzidine tetrahydrochloride and 0.01 % hydrogen peroxide. Control sections were incubated without primary antiserum.

Results

Experiment 1

The results are summarized in Table I. In the vehicle-treated animals, the water content in the brain significantly increased in the ipsilateral hemisphere, being most severe in the caudate-putamen (8.8 % increase; $p < 0.01$ vs sham-operation) and the MCA cortex (6.8 %; $p < 0.01$) and moderate in the ACA cortex (2.2 %; $p < 0.05$). We also observed a slight but significant increase in water content in the contralateral caudate-putamen. Treatment with F-0401 (both 1 and 10 mg/kg) resulted in significant reductions in water content in the MCA and ACA cortices and caudate-putamen of the ipsilateral hemisphere and in the caudate-putamen of the contralateral hemisphere.

Experiment 2

The effects of F-0401 on the extravasation of Evans blue dye in the brain after ischemia and reperfusion, are shown in Table II. There was no region showing visible dye extravasation in sham-operated brains. The brains subjected to 2 hr of ischemia followed by 2 hr of reperfusion, showed a dramatical extravasation, particularly in the cortex, striatum, thalamus and hippocampus. A significant reduction of the extravasated dye was observed in F-0401-treated animals.

Experiment 3

Microphotographs of the hippocampus are shown in Fig. 2 and the neuronal density of the hippocampal CA1 subfield is shown in Table III. Three and 5 min of ischemia destroyed 56 % and 95 % of CA1 neurons 7 days after ischemia, respectively. The neuronal density of the hippocampal CA1 subfield of the animals treated with F-0401 (1 and 10 mg/kg) was significantly higher than that of the vehicle-treated groups after both 3 and 5 min of ischemia.

TABLE I

Water content (%) of various brain regions 24 hr after 1 hr of right middle cerebral artery occlusion in the rat

	n	MCA cortex		ACA cortex		Caudate-putamen	
		Right	Left	Right	Left	Right	Left
Sham-operated	8	79.0 ± 0.3**	78.9 ± 0.2	78.7 ± 0.4	78.6 ± 0.3	76.7 ± 1.1**	76.1 ± 1.3*
Vehicle-treated	13	85.8 ± 1.0	78.9 ± 0.4	81.9 ± 1.5	78.7 ± 0.4	85.5 ± 1.2	77.2 ± 1.3
F-0401 1 mg/kg	8	84.0 ± 1.9*	78.7 ± 0.4	80.2 ± 1.3*	78.7 ± 0.3	83.9 ± 1.2*	76.0 ± 0.9*
F-0401 10 mg/kg	13	83.6 ± 1.9*	78.3 ± 0.8*	80.1 ± 1.3**	78.3 ± 0.5	84.1 ± 1.6*	75.9 ± 1.2**

MCA and ACA: Middle and anterior cerebral arteries, respectively.

Values are expressed as means ± S.D.

* $p < 0.05$, ** $p < 0.01$: vs vehicle-treated animals (Williams-Wilcoxon multiple rank sum test).

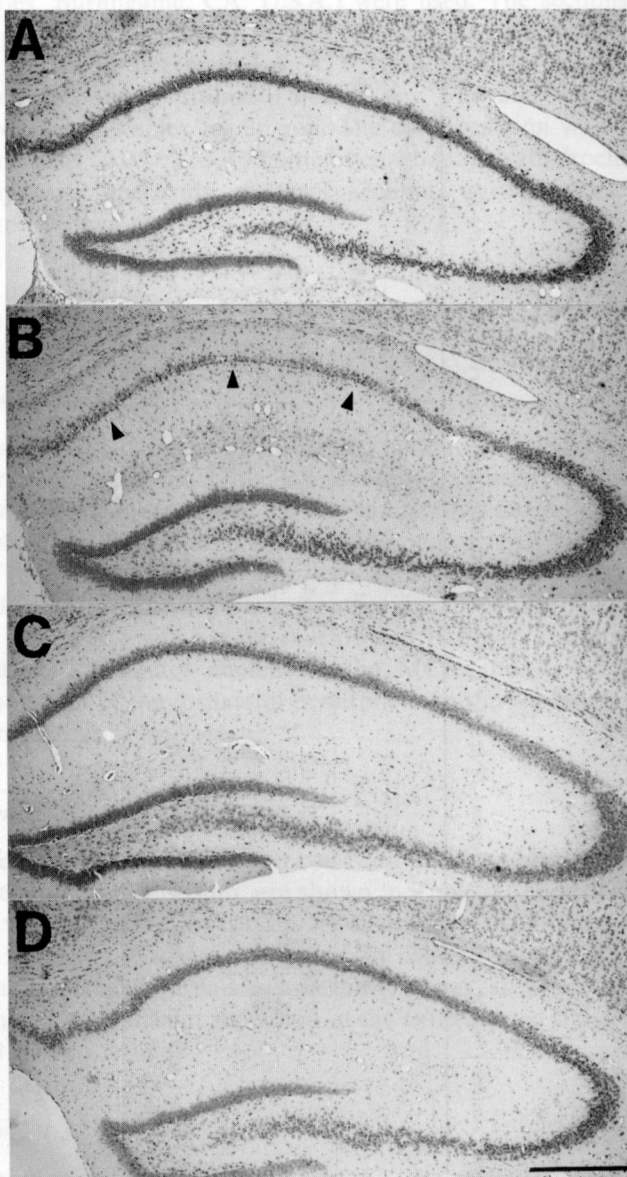


FIG. 2

Microphotographs of the gerbil hippocampus after 3 min of forebrain ischemia.

A: Sham-operated; B: vehicle-treated; C: F-0401 (1 mg/kg); D: F-0401 (10 mg/kg).

Note CA1 neuronal loss in B but not in C and D.

Cresyl violet stain. Calibration bar: 0.5 mm.

TABLE II

Degree of Evans blue dye extravasation in the brain following 2 hr of bilateral carotid artery occlusion and 2 hr of reperfusion in the rat

	n	Evans blue dye extravasation
Sham-operated	6	$0 \pm 0^{**}$
Vehicle-treated	8	1.6 ± 1.2
F-0401 1 mg/kg	6	$0.2 \pm 0.1^*$
F-0401 10 mg/kg	6	$0.3 \pm 0.2^*$

The degree of dye extravasation was semiquantitatively graded as 0 (none), 1 (mild), 2 (moderate) and 3 (severe).

Values are expressed as means $\pm$ S.D.

* $p < 0.05$, ** $p < 0.01$: vs vehicle (Williams-Wilcoxon multiple rank sum test).

The immunohistochemical localization of MAP2 appeared to be preferentially localized in the neuronal perikarya and dendrites (Fig. 3). The MAP2 immunoreactivity in the CA1 region of the hippocampus was lost following 3 min of ischemia. However, the immunoreactivity was preserved in the F-0401-treated animals (Fig. 3).

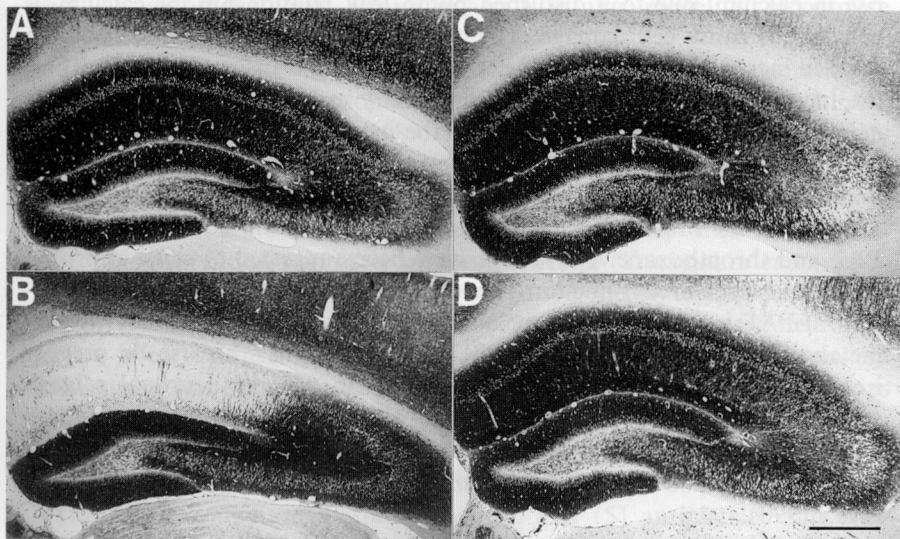


FIG. 3

Immunostaining against microtubule-associated protein-2 (MAP2) in the hippocampus after 3 min of ischemia.

A: Sham-operated; B: vehicle-treated; C: F-0401 (1 mg/kg); D: F-0401 (10 mg/kg).

Note loss of MAP2 immunoreactivity in CA1 in B.

Calibration bar: 0.5 mm.

TABLE III

Neuronal density (/mm) of the CA1 subfield of the hippocampus after 3 min and 5 min of forebrain ischemia in the gerbil

	3 min of ischemia		5 min of ischemia	
	n	Neuronal density	n	Neuronal density
Sham-operated	3	218 ± 8.3**	3	218 ± 8.3**
Vehicle-treated	8	95 ± 19.6	9	10 ± 2.6
F-0401 1 mg/kg	6	163 ± 3.3**	10	20 ± 6.1**
F-0401 10 mg/kg	6	173 ± 5.6**	8	47 ± 66.8**

Values are expressed as means ± S.D.

** $p < 0.01$: vs vehicle (Williams-Wilcoxon multiple rank sum test).

Discussion

The mechanisms of ischemic brain damage have not been definitively elucidated, but cumulative evidence points to an increase in the intracellular calcium concentration (Siesjö, 1992a, 1992b). A nonphysiological rise in calcium, due to a disturbed pump/leak relationship for calcium, is believed to cause cell damage. The calcium content increases, preceding a marked increase in neuronal cell necrosis following transient ischemia. Thus, the increase in calcium content is a primary event leading to irreversible neuronal damage and is indicative of the extent of cell death (Deshpande *et al.*, 1987). Therefore, the ischemic lesions may be ameliorated by agents that improve the pump/leak relationship for calcium (Siesjö, 1992a, 1992b). On the other hand, chemical mediators of brain damage, including PAF and thromboxane A_2 (TXA₂), have been suggested to cause neuronal and microvascular dysfunction (Feuerstein *et al.*, 1990). Intracarotid infusion of PAF into normal rats produces cerebral hypoperfusion and hypermetabolism (Kochanek *et al.*, 1990). PAF antagonists reverse or prevent ischemic brain injury and have the ability to achieve neuronal salvage (Feuerstein *et al.*, 1990). Pretreatment with PAF antagonists improves impaired microcirculation and reduces edema and BBB permeability as well as neutrophil accumulation following focal brain injury (Frerichs *et al.*, 1990).

In this study, we showed that F-0401 reduced brain edema and BBB permeability, and prevented delayed neuronal death of hippocampal neurons in the rat and gerbil models of cerebral ischemia. The mechanism of the attenuation of ischemic brain damage may be attributed both to a calcium antagonistic action and to a PAF antagonistic action of F-0401, because earlier studies have demonstrated that calcium antagonists and

PAF antagonists, given alone, ameliorate brain damage in cerebral ischemia models (Spinnewyn *et al.*, 1987; Abe *et al.*, 1988; Hara *et al.*, 1990, 1991; Sauter and Rudin, 1990; Gilboe *et al.*, 1991; Bielenberg *et al.*, 1992).

In the MCA occlusion model, brain edema was most severe in the ischemic core (the caudate-putamen and the MCA cortex) and was moderate in the surrounding area (the ACA cortex), as reported previously (Koizumi *et al.*, 1986; Nagasawa and Kogure, 1989). F-0401 significantly reduced by 1.4–2.2 % the water content in these three brain areas as compared to vehicle-treated animals. In this study, we observed a slight increase in water content in the contralateral hemisphere. This contralateral edema, which was ameliorated by F-0401 treatment, may be induced by a secondary circulatory disturbance due to mass effects of the primary lesion.

The vasogenic edema, which develops after reperfusion of the cerebral blood flow (Ito *et al.*, 1979; Sadoshima *et al.*, 1983), and the dysfunction of BBB, can be detected by extravasation of conventional BBB indicators such as Evans blue (Ito *et al.*, 1976; Kogure *et al.*, 1981; Sadoshima *et al.*, 1983; Bhakoo *et al.*, 1984). In the present study, a large and significant decrease of extravasation of Evans blue was observed in F-0401-treated animals as compared to vehicle-treated animals. This protective effect may be due to the fact that F-0401 increased cerebral blood flow and protected vascular endothelial cells during ischemia in this model of incomplete cerebral ischemia.

Hippocampal CA1 pyramidal neurons are selectively vulnerable to ischemia, a 3–5 min duration of which causes severe neuronal damage (Kirino, 1982; Araki *et al.*, 1989; Kato *et al.*, 1991). The present study demonstrated that treatment with F-0401 prevents delayed neuronal death of the hippocampal CA1 pyramidal neurons following 3 and 5 min of ischemia. MAP2, which is a marker of dendrites and can detect morphological damage in early cerebral ischemia (Yanagihara *et al.*, 1985; Matsumoto *et al.*, 1987; Kitagawa *et al.*, 1989), clearly demonstrated that the CA1 neurons are protected by F-0401. Thus, F-0401 has a neuroprotective effect against hippocampal neuronal damage following transient ischemia, as has been demonstrated for calcium and PAF antagonists (Alps *et al.*, 1988; Lindsberg *et al.*, 1991).

The pharmacological actions of F-0401 are (1) a calcium antagonistic action, (2) a PAF antagonistic action, (3) dilatation of cerebral arteries, (4) a TXA₂ synthetase inhibitory action, and (5) antiplatelet and anti-thrombotic effects (Iida *et al.*, 1991; Kase *et al.*, 1993, 1993b). Although the pharmacological profile of F-0401, summarized in these five points, may be brought about by the first two activities, F-0401 is expected to act on multiple aspects of ischemic brain damage.

Although two doses of F-0401 may be insufficient to examine correct dose-effect relationships, the effects of the drug were studied in three

different experimental ischemia models and the results were largely dose-dependent. Furthermore, comparative measurements, using a calcium channel blocker or a PAF antagonist alone or in combination, may be necessary to really prove that a drug with a dual effect has priority over drugs representing either a pharmacological group or a combination of drugs with two types of pharmacological activity. Therefore, we cannot conclude that a therapeutic agent with multiple actions, such as F-0401, may be more promising in the treatment of ischemic stroke than a therapeutic agent with a single effect.

However, we used three different cerebral ischemia models in this study to test the effects of F-0401. The present findings showed that F-0401 ameliorates ischemic brain edema and disruption of BBB, and prevents delayed neuronal death of hippocampal neurons. Thus, this study demonstrated that F-0401 may be a powerful candidate as a therapeutic agent in the treatment of acute stroke in man.

Acknowledgments—F-0401 was kindly donated by Fujirebio Inc. (Tokyo, Japan). We are very grateful to Dr. Y. Sekine, Fujirebio Inc., for his helpful comments and advice.

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